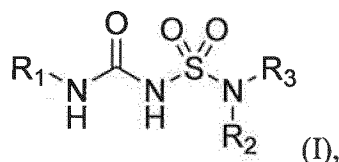


CLAIMS:

1. A compound of Formula (I):



or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein:

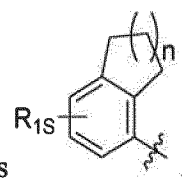
R₁ is C₃-C₁₆ cycloalkyl, or C₅-C₁₀ aryl, wherein the C₃-C₈ monocyclic cycloalkyl, polycyclic cycloalkyl, or C₅-C₆ aryl is optionally substituted by one or more **R_{1S}**; wherein each **R_{1S}** is independently C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₁-C₆ alkoxy, C₁-C₆ haloalkoxy, or halo;

R₂ is -(CX₂X₂)_n-**R_{2S}**, wherein n is 0, 1, or 2, and each **X₂** is independently H, C₁-C₆ alkyl, C₂-C₆ alkenyl, or C₂-C₆ alkynyl, wherein the C₁-C₆ alkyl, C₂-C₆ alkenyl, or C₂-C₆ alkynyl is optionally substituted with one or more halo, -CN, -OH, -O(C₁-C₆ alkyl), -NH₂, -NH(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, or oxo;

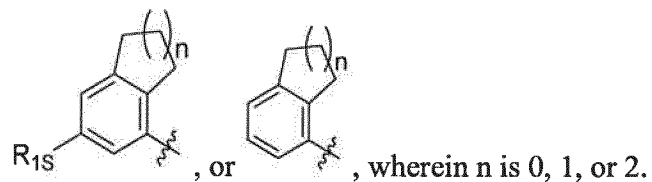
R_{2S} is 4- to 8-membered heterocycloalkyl optionally substituted with one or more C₁-C₆ alkyl, C₂-C₆ alkenyl, C₂-C₆ alkynyl, C₁-C₆ haloalkyl, halo, -CN, -OH, -O(C₁-C₆ alkyl), -NH₂, -NH(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, or oxo; and

R₃ is 7- to 12-membered heterocycloalkyl or 5- or 6-membered heteroaryl optionally substituted with one or more **R_{3S}**, wherein each **R_{3S}** is independently C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, halo, or C₃-C₈ heterocycloalkyl wherein the C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, or C₃-C₈ heterocycloalkyl is optionally substituted with -O(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, halo, or -CN.

2. The compound of any one of the preceding claims, wherein **R₁** is C₉-C₁₀ bicyclic cycloalkyl.

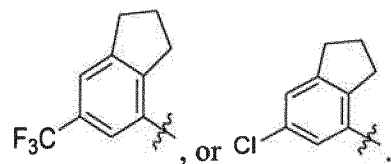


3. The compound of any one of the preceding claims, wherein **R₁** is

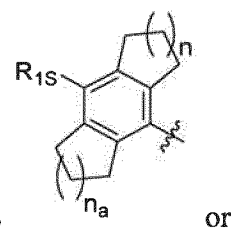


4. The compound of any one of the preceding claims, wherein **R₁** is

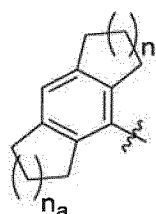
5. The compound of any one of the preceding claims, wherein **R₁** is



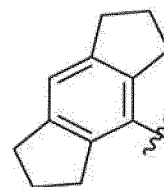
6. The compound of any one of the preceding claims, wherein **R₁** is C₁₂-C₁₆ tricyclic cycloalkyl.



7. The compound of any one of the preceding claims, wherein **R₁** is



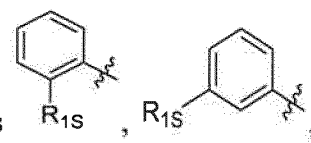
, wherein **n** and **n_a** each independently are 0, 1, 2, or 3.



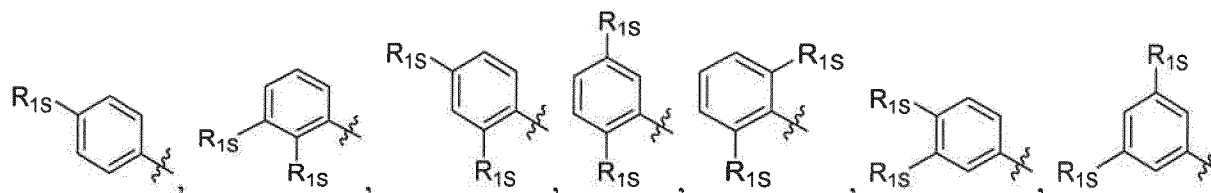
8. The compound of any one of the preceding claims, wherein **R₁** is

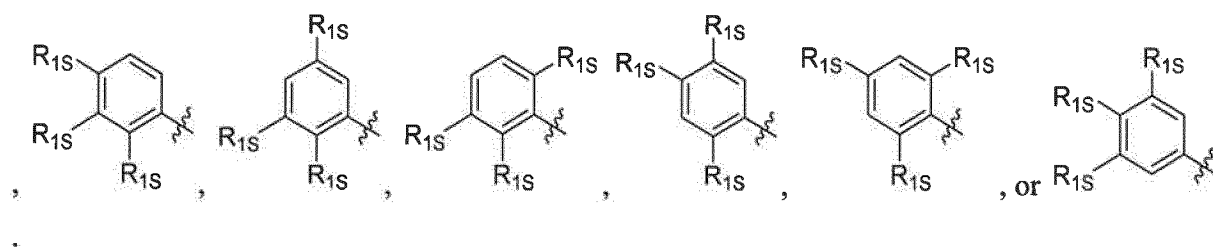
9. The compound of any one of the preceding claims, wherein, **R₁** is C₅-C₁₀ aryl optionally substituted by one or more **R_{1S}**.

10. The compound of any one of the preceding claims, wherein **R₁** is phenyl is substituted by one, two, or three **R_{1S}**.

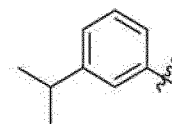


11. The compound of any one of the preceding claims, wherein **R₁** is

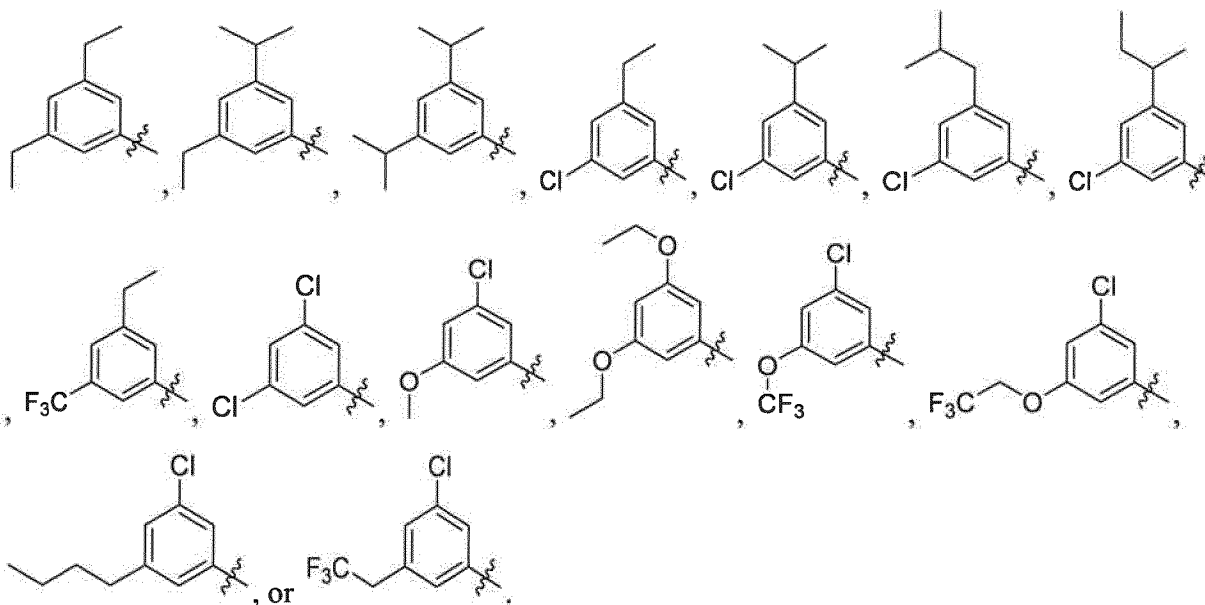
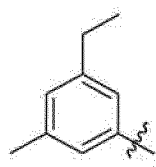


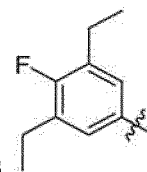


12. The compound of any one of the preceding claims, wherein **R₁** is

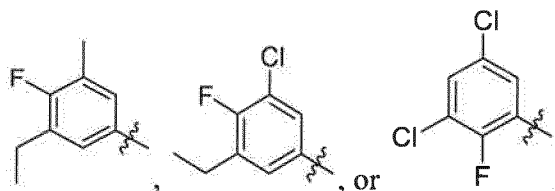


13. The compound of any one of the preceding claims, wherein **R₁** is

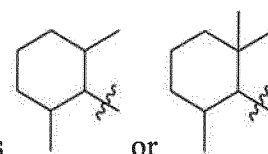




14. The compound of any one of the preceding claims, wherein **R**₁ is



15. The compound of any one of the preceding claims, wherein **R**₁ is cyclopentyl, cyclohexyl, or cycloheptyl, wherein the cyclopentyl, cyclohexyl, or cycloheptyl is optionally substituted by one or more **R**_{1S}.



16. The compound of any one of the preceding claims, wherein **R**₁ is

17. The compound of any one of the preceding claims, wherein at least one **R**_{1S} is methyl, ethyl, isopropyl, isobutyl, secbutyl, methoxy, ethoxy, -CF₃, -OCF₃, -OCH₂CF₃, F, or Cl.

18. The compound of any one of the preceding claims, wherein **R**₂ is -**R**_{2S}.

19. The compound of any one of the preceding claims, wherein **R**₂ is -(CX₂X₂)-**R**_{2S}.

20. The compound of any one of the preceding claims, wherein **R**₂ is -(CX₂X₂)₂-**R**_{2S}.

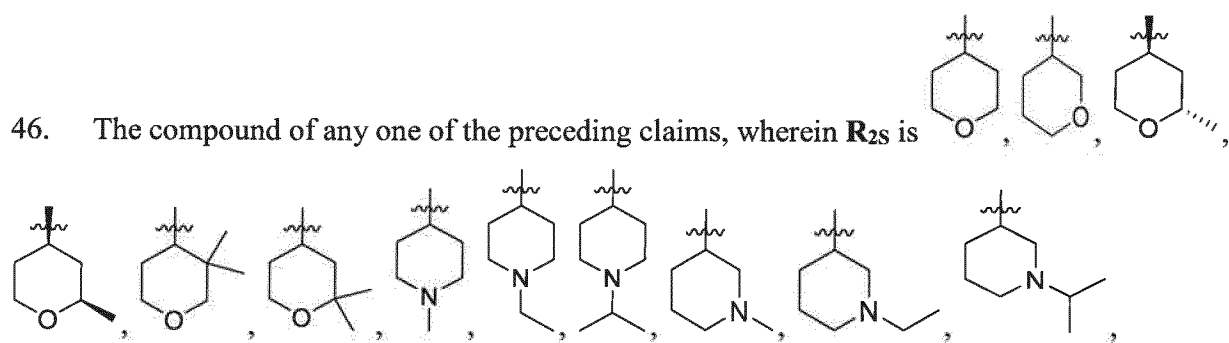
21. The compound of any one of the preceding claims, wherein each X₂ is H.

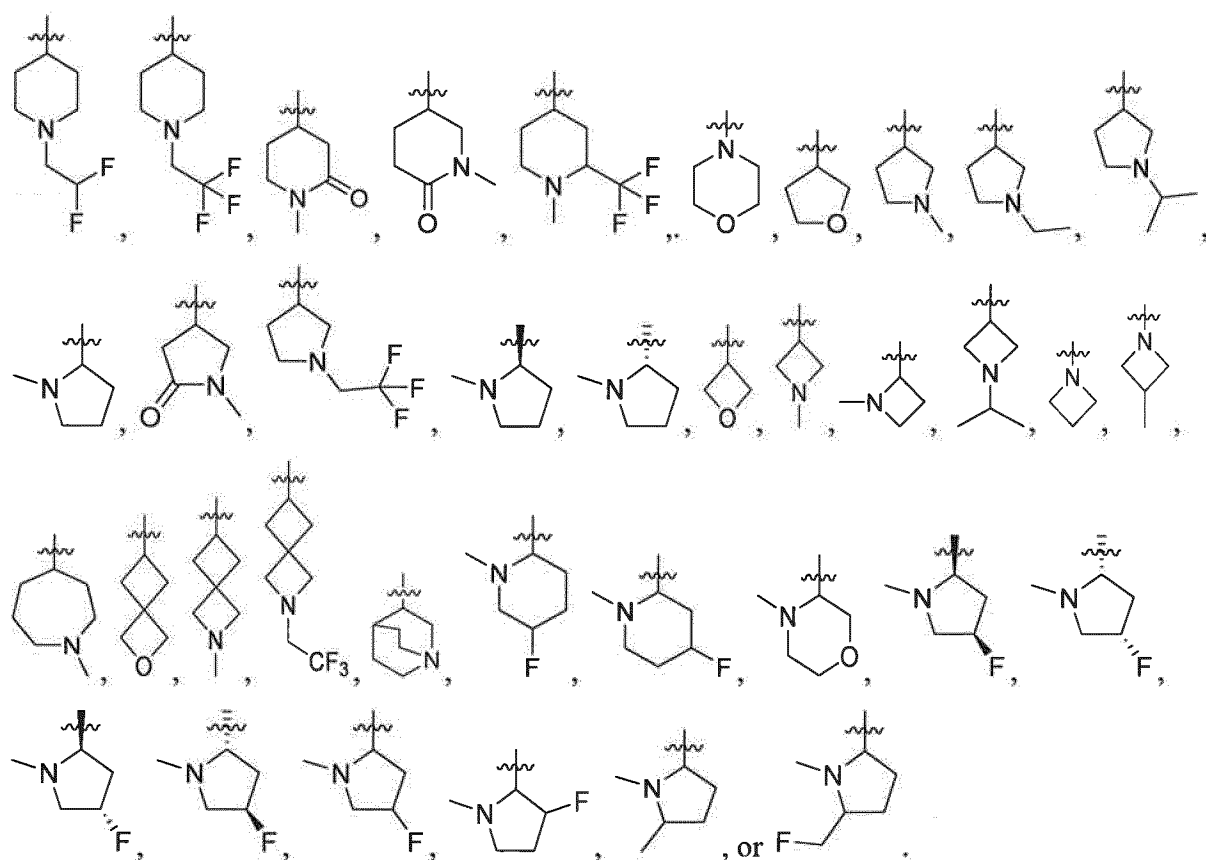
22. In some embodiments, at least one X₂ is C₁-C₆ alkyl, C₂-C₆ alkenyl, or C₂-C₆ alkynyl, wherein the C₁-C₆ alkyl, C₂-C₆ alkenyl, or C₂-C₆ alkynyl is optionally substituted with one or more halo, -CN, -OH, -O(C₁-C₆ alkyl), -NH₂, -NH(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, or oxo.

23. The compound of any one of the preceding claims, wherein **R_{2S}** is 4- to 8-membered heterocycloalkyl optionally substituted with one or more C₁-C₆ alkyl, C₂-C₆ alkenyl, C₂-C₆ alkynyl, C₁-C₆ haloalkyl, halo, -CN, -OH, -O(C₁-C₆ alkyl), -NH₂, -NH(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, or oxo.
24. The compound of any one of the preceding claims, wherein **R_{2S}** is 4- to 8-membered heterocycloalkyl.
25. The compound of any one of the preceding claims, wherein **R_{2S}** is 5- to 8-membered heterocycloalkyl.
26. The compound of any one of the preceding claims, wherein **R_{2S}** is 5- to 7-membered heterocycloalkyl.
27. The compound of any one of the preceding claims, wherein **R_{2S}** is 5- to 6-membered heterocycloalkyl.
28. The compound of any one of the preceding claims, wherein **R_{2S}** is 6-membered heterocycloalkyl optionally substituted with one or more C₁-C₆ alkyl, C₂-C₆ alkenyl, C₂-C₆ alkynyl, C₁-C₆ haloalkyl, halo, -CN, -OH, -O(C₁-C₆ alkyl), -NH₂, -NH(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, or oxo.
29. The compound of any one of the preceding claims, wherein **R_{2S}** is tetrahydropyranyl optionally substituted with one or more C₁-C₆ alkyl, C₁-C₆ haloalkyl, or oxo.
30. The compound of any one of the preceding claims, wherein **R_{2S}** is piperidinyl optionally substituted with one or more C₁-C₆ alkyl, C₁-C₆ haloalkyl, or oxo.
31. The compound of any one of the preceding claims, wherein **R_{2S}** is morpholinyl optionally substituted with one or more C₁-C₆ alkyl, C₁-C₆ haloalkyl, or oxo.

32. The compound of any one of the preceding claims, wherein **R_{2S}** is 5-membered heterocycloalkyl optionally substituted with one or more C₁-C₆ alkyl, C₂-C₆ alkenyl, C₂-C₆ alkynyl, C₁-C₆ haloalkyl, halo, -CN, -OH, -O(C₁-C₆ alkyl), -NH₂, -NH(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, or oxo.
33. The compound of any one of the preceding claims, wherein **R_{2S}** is tetrahydrofuranyl optionally substituted with one or more C₁-C₆ alkyl, C₁-C₆ haloalkyl, or oxo.
34. The compound of any one of the preceding claims, wherein **R_{2S}** is pyrrolidinyl optionally substituted with one or more C₁-C₆ alkyl, C₁-C₆ haloalkyl, or oxo.
35. The compound of any one of the preceding claims, wherein **R_{2S}** is 4-membered heterocycloalkyl optionally substituted with one or more C₁-C₆ alkyl, C₂-C₆ alkenyl, C₂-C₆ alkynyl, C₁-C₆ haloalkyl, halo, -CN, -OH, -O(C₁-C₆ alkyl), -NH₂, -NH(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, or oxo.
36. The compound of any one of the preceding claims, wherein **R_{2S}** is oxetanyl optionally substituted with one or more C₁-C₆ alkyl, C₁-C₆ haloalkyl, or oxo.
37. The compound of any one of the preceding claims, wherein **R_{2S}** is azetidiny optionally substituted with one or more C₁-C₆ alkyl, C₁-C₆ haloalkyl, or oxo.
38. The compound of any one of the preceding claims, wherein **R_{2S}** is 7-membered heterocycloalkyl optionally substituted with one or more C₁-C₆ alkyl, C₂-C₆ alkenyl, C₂-C₆ alkynyl, C₁-C₆ haloalkyl, halo, -CN, -OH, -O(C₁-C₆ alkyl), -NH₂, -NH(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, or oxo.
39. The compound of any one of the preceding claims, wherein **R_{2S}** is azepanyl optionally substituted with one or more C₁-C₆ alkyl, C₁-C₆ haloalkyl, or oxo.

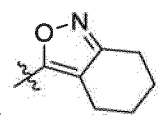
40. The compound of any one of the preceding claims, wherein **R_{2S}** is 8-membered heterocycloalkyl optionally substituted with one or more C₁-C₆ alkyl, C₂-C₆ alkenyl, C₂-C₆ alkynyl, C₁-C₆ haloalkyl, halo, -CN, -OH, -O(C₁-C₆ alkyl), -NH₂, -NH(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, or oxo.
41. The compound of any one of the preceding claims, wherein **R_{2S}** is spiro 7-membered heterocycloalkyl optionally substituted with one or more C₁-C₆ alkyl, C₁-C₆ haloalkyl, or oxo.
42. The compound of any one of the preceding claims, wherein **R_{2S}** is 2-azaspiro[3.3]heptane optionally substituted with one or more C₁-C₆ alkyl, C₁-C₆ haloalkyl, or oxo.
43. The compound of any one of the preceding claims, wherein **R_{2S}** is 2-oxaspiro[3.3]heptane optionally substituted with one or more C₁-C₆ alkyl, C₁-C₆ haloalkyl, or oxo.
44. The compound of any one of the preceding claims, wherein **R_{2S}** is bridged heterocycloalkyl optionally substituted with one or more C₁-C₆ alkyl, C₂-C₆ alkenyl, C₂-C₆ alkynyl, C₁-C₆ haloalkyl, halo, -CN, -OH, -O(C₁-C₆ alkyl), -NH₂, -NH(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, or oxo.
45. The compound of any one of the preceding claims, wherein **R_{2S}** is bridged quinuclidinyl optionally substituted with one or more C₁-C₆ alkyl, C₁-C₆ haloalkyl, or oxo.





47. The compound of any one of the preceding claims, wherein **R**₃ is 7- to 12-membered heterocycloalkyl optionally substituted with one or more **R**_{3s}.

48. The compound of any one of the preceding claims, wherein **R**₃ is



49. The compound of any one of the preceding claims, wherein **R**₃ is 5- or 6-membered heteroaryl optionally substituted with one or more **R**_{3s}, wherein each **R**_{3s} is independently C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, halo, or C₃-C₈ heterocycloalkyl wherein the C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, or C₃-C₈ heterocycloalkyl is optionally substituted with -O(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, halo, or -CN.

50. The compound of any one of the preceding claims, wherein **R**₃ is 5-membered heteroaryl optionally substituted with one or more **R**_{3s}, wherein each **R**_{3s} is independently C₁-C₆ alkyl, C₁-

C₆ haloalkyl, C₃-C₈ cycloalkyl, halo, or C₃-C₈ heterocycloalkyl wherein the C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, or C₃-C₈ heterocycloalkyl is optionally substituted with -O(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, halo, or -CN.

51. The compound of any one of the preceding claims, wherein **R₃** is 6-membered heteroaryl optionally substituted with one or more **R_{3S}**, wherein each **R_{3S}** is independently C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, halo, or C₃-C₈ heterocycloalkyl wherein the C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, or C₃-C₈ heterocycloalkyl is optionally substituted with -O(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, halo, or -CN.

52. The compound of any one of the preceding claims, wherein **R₃** is pyrrolyl optionally substituted by one or more **R_{3S}**.

53. The compound of any one of the preceding claims, wherein **R₃** is pyrazolyl optionally substituted by one or more **R_{3S}**.

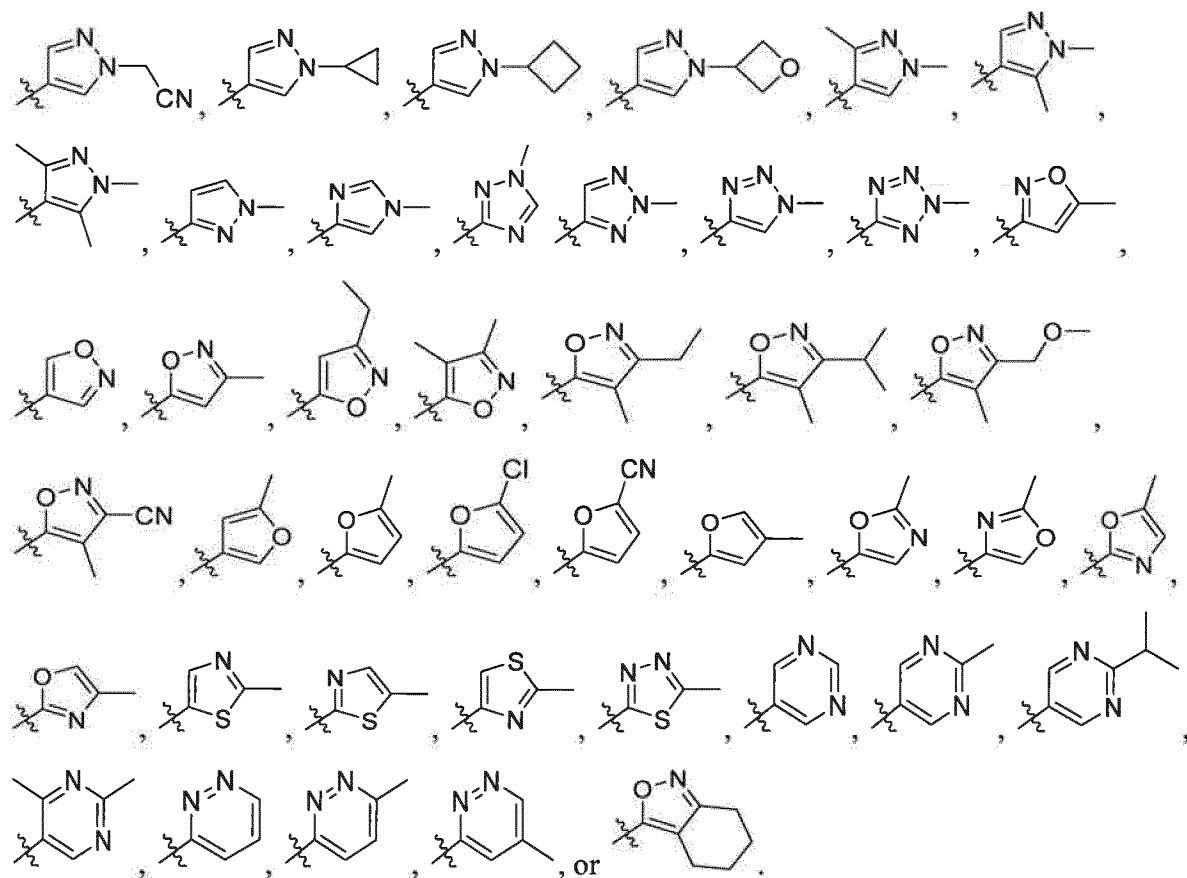
54. The compound of any one of the preceding claims, wherein **R₃** is imidazolyl optionally substituted by one or more **R_{3S}**.

55. The compound of any one of the preceding claims, wherein **R₃** is triazolyl optionally substituted by one or more **R_{3S}**.

56. The compound of any one of the preceding claims, wherein **R₃** is tetrazolyl optionally substituted by one or more **R_{3S}**.

57. The compound of any one of the preceding claims, wherein **R₃** is isoxazolyl optionally substituted by one or more **R_{3S}**.

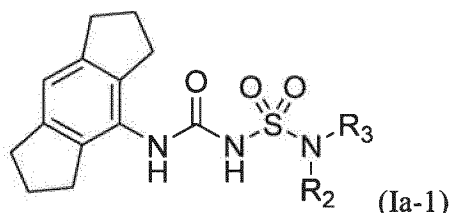
58. The compound of any one of the preceding claims, wherein **R₃** is furanyl optionally substituted by one or more **R_{3S}**.



67. The compound of any one of the preceding claims, wherein at least one **R**₃₈ is C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, halo, or C₃-C₈ heterocycloalkyl wherein the C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, or C₃-C₈ heterocycloalkyl is optionally substituted with -O(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, halo, or -CN.

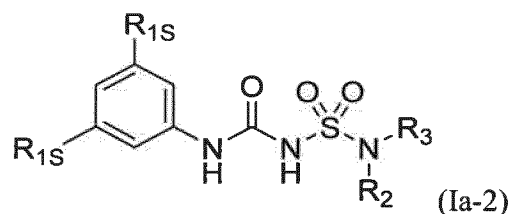
68. The compound of any one of the preceding claims, wherein at least one R_{3S} is , , or , methyl, ethyl, isopropyl, $-CH_2OCH_3$, $-CH_2CF_3$, $-CH_2CH_2OCH_3$, $-CH_2CN$, CH_2CF_2 , or $-CH_2CH_2N(CH_3)_2$.

69. The compound of any one of the preceding claims, wherein the compound is of Formula (Ia-1):



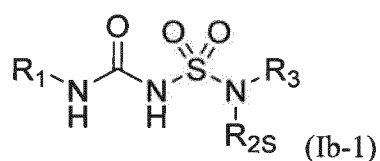
or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R₂** and **R₃** are as described herein.

70. The compound of any one of the preceding claims, wherein the compound is of Formula (Ia-2):



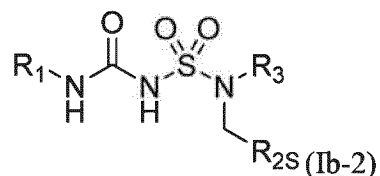
or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R_{1S}**, **R₂** and **R₃** are as described herein.

71. The compound of any one of the preceding claims, wherein the compound is of Formula (Ib-1):



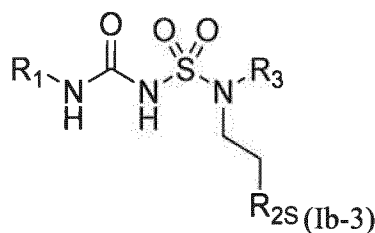
or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R₁**, **R_{2S}**, and **R₃** are as described herein.

72. The compound of any one of the preceding claims, wherein the compound is of Formula (Ib-2):



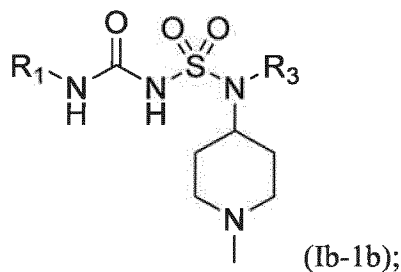
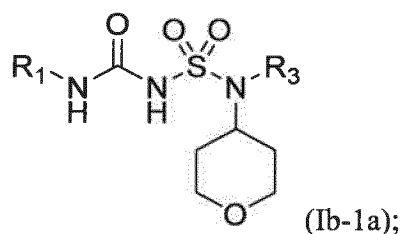
or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R₁**, **R_{2S}**, and **R₃** are as described herein.

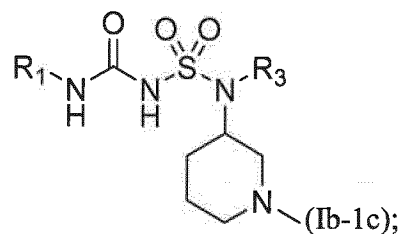
73. The compound of any one of the preceding claims, wherein the compound is of Formula (Ib-3):



or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R₁**, **R_{2S}**, and **R₃** are as described herein.

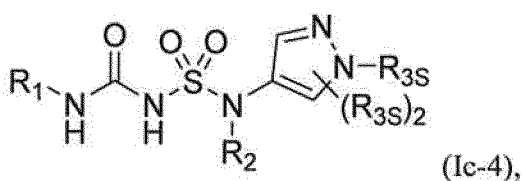
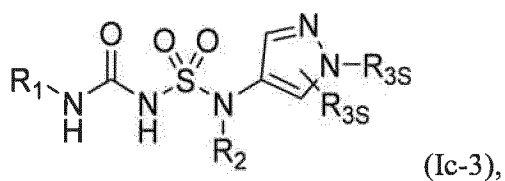
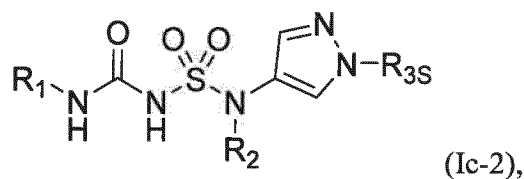
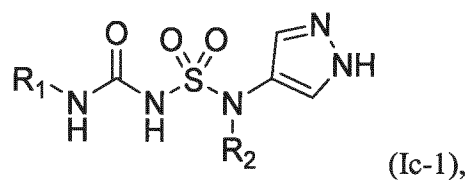
74. The compound of any one of the preceding claims, wherein the compound is of any one of Formulae (Ib-1a), (Ib-1b), or (Ib-1c):





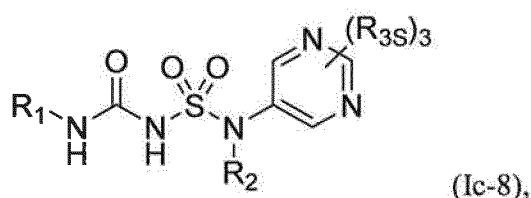
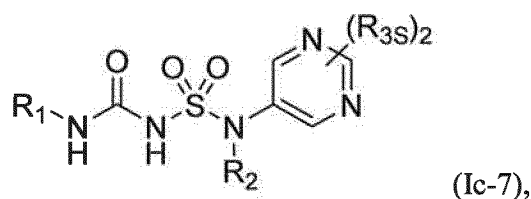
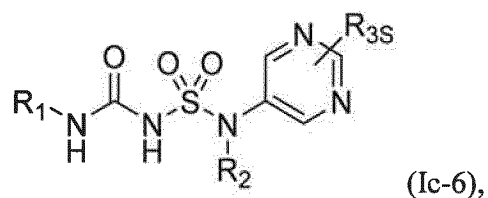
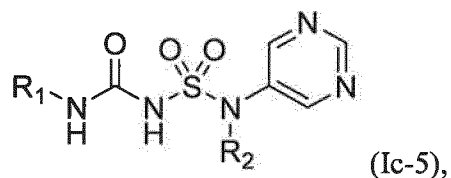
or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R₁** and **R₃** are as described herein.

75. The compound of any one of the preceding claims, wherein the compound is of Formula (Ic-1), (Ic-2), (Ic-3), or (Ic-4):



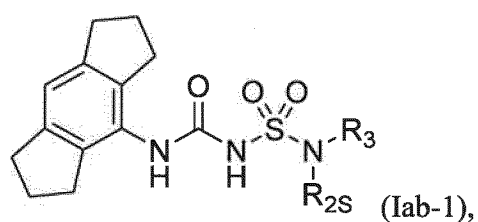
or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R₁**, **R₂** and **R_{3S}**, are as described herein.

76. The compound of any one of the preceding claims, wherein the compound is of Formula (Ic-5), (Ic-6), (Ic-7), or (Ic-8):



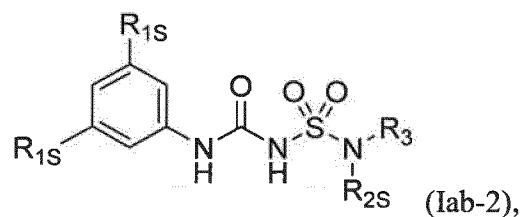
or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R**₁, **R**₂ and **R**_{3S}, are as described herein.

77. The compound of any one of the preceding claims, wherein the compound is of Formula (Iab-1):



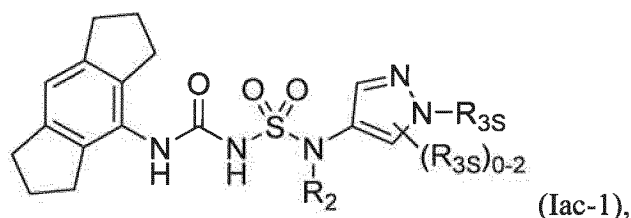
or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R**_{2S} and **R**₃ are as described herein.

78. The compound of any one of the preceding claims, wherein the compound is of Formula (Iab-2):



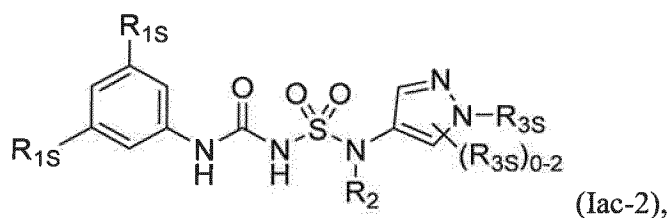
or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R_{1S}**, **R_{2S}**, and **R₃** are as described herein.

79. The compound of any one of the preceding claims, wherein the compound is of Formula (Iac-1):



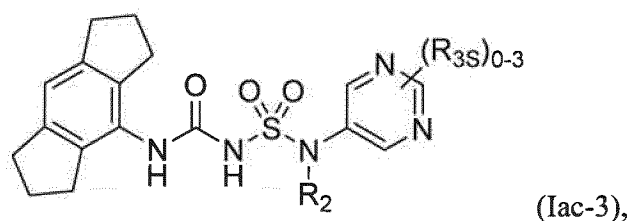
or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R₂** and **R_{3S}** are as described herein.

80. The compound of any one of the preceding claims, wherein the compound is of Formula (Iac-2):



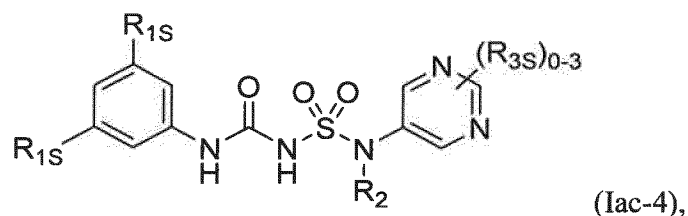
or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R_{1S}**, **R₂**, and **R_{3S}** are as described herein.

81. The compound of any one of the preceding claims, wherein the compound is of Formula (Iac-3):



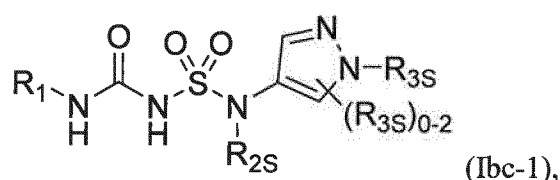
or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R₂** and **R_{3s}** are as described herein.

82. The compound of any one of the preceding claims, wherein the compound is of Formula (Iac-4):



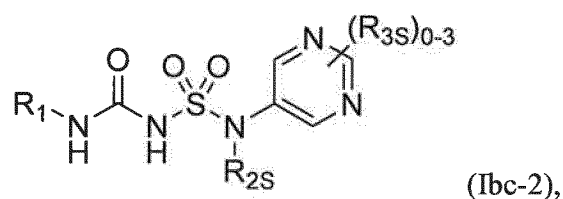
or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R_{1s}**, **R₂**, and **R_{3s}** are as described herein.

83. The compound of any one of the preceding claims, wherein the compound is of Formula (Ibc-1):



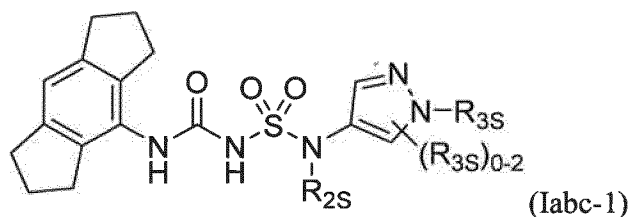
or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R₁**, **R_{2s}**, and **R_{3s}** are as described herein.

84. The compound of any one of the preceding claims, wherein the compound is of Formula (Ibc-2):



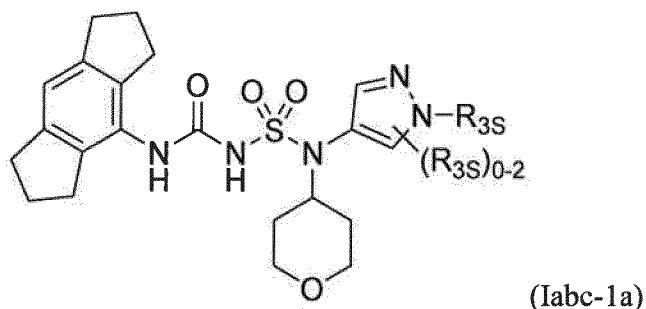
or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein R_1 , R_{2s} , and R_{3s} are as described herein.

85. The compound of any one of the preceding claims, wherein the compound is of Formula (Iabc-1):



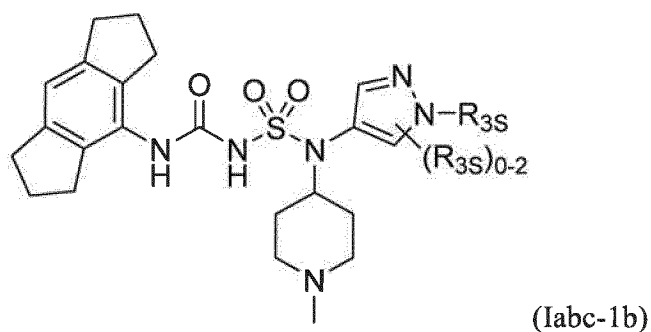
or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein R_{2s} and R_{3s} are as described herein.

86. The compound of any one of the preceding claims, wherein the compound is of Formula (Iabc-1a):



or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein R_{3s} is as described herein.

87. The compound of any one of the preceding claims, wherein the compound is of Formula (Iabc-1b):



or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R_{3s}** is as described herein.

88. The compound of any one of the claims 1-87, being selected from Compound Nos. 1-166 and prodrugs and pharmaceutically acceptable salts thereof.
89. The compound of any one of claims 1-87, being selected from Compound Nos. 1-166 and pharmaceutically acceptable salts thereof.
90. The compound of any one of claims 1-87, being selected from Compound Nos. 1-166.
91. The compound of any one of the claims 1-87, being selected from Compound Nos. 1A-166A and prodrugs and pharmaceutically acceptable salts thereof.
92. The compound of any one of claims 1-87, being selected from Compound Nos. 1A-166A and pharmaceutically acceptable salts thereof.
93. The compound of any one of claims 1-87, being selected from Compound Nos. 1A-166A.
94. A compound being an isotopic derivative of the compound of any one of claims 1-93.
95. The compound of claim 94, being a deuterium labeled compound.
96. The compound of claim 94, being a deuterium labeled compound of any one of Compound Nos. 1-166 and prodrugs and pharmaceutically acceptable salts thereof.
97. The compound of claim 94, being a deuterium labeled compound of any one of Compound Nos. 1-166.
98. The compound of claim 94, being a deuterium labeled compound of any one of Compound Nos. 1A-166A and prodrugs and pharmaceutically acceptable salts thereof.

99. The compound of claim 94, being a deuterium labeled compound of any one of Compound Nos. 1A-166A.
100. A compound obtainable by, or obtained by, a method described herein; optionally, the method comprises one or more steps described in Schemes 1-2.
101. A compound, by an intermediate obtained by a method for preparing the compound of any one of claims 1-93; optionally, the intermediate is selected from the intermediates described in Examples 1-166.
102. A pharmaceutical composition comprising the compound of any one of claims 1-101 or a pharmaceutically acceptable salt thereof, and a pharmaceutically acceptable diluent or carrier.
103. The pharmaceutical composition of claim 102, wherein the compound is selected from Compound Nos. 1-166.
104. The pharmaceutical composition of claim 102, wherein the compound is selected from Compound Nos. 1A-166A.
105. A method of inhibiting inflammasome activity, comprising contacting a cell with an effective amount of the compound of any one of claims 1-101 or a pharmaceutically acceptable salt thereof; optionally, the inflammasome is NLRP3 inflammasome, and the activity is *in vitro* or *in vivo*.
106. A method of treating or preventing a disease or disorder in a subject in need thereof, comprising administering to the subject a therapeutically effective amount of the compound of any one of claims 1-101 or a pharmaceutically acceptable salt thereof, or the pharmaceutical composition of any one of claims 102-104.

107. The method of claim 106, wherein the disease or disorder is associated with an implicated inflammasome activity; optionally, the disease or disorder is a disease or disorder in which inflammasome activity is implicated.

108. The method of claim 106 or claim 107, wherein the disease or disorder is an inflammatory disorder, an autoinflammatory disorder, an autoimmune disorder, a neurodegenerative disease, or cancer.

109. The method of any one of claims 106-108, wherein the disease or disorder is an inflammatory disorder, an autoinflammatory disorder or an autoimmune disorder; optionally, the disease or disorder is selected from cryopyrin-associated auto-inflammatory syndrome (CAPS; e.g., familial cold autoinflammatory syndrome (FCAS), Muckle-Wells syndrome (MWS), chronic infantile neurological cutaneous and articular (CINCA) syndrome/ neonatal-onset multisystem inflammatory disease (NOMID)), familial Mediterranean fever (FMF), nonalcoholic fatty liver disease (NAFLD), non-alcoholic steatohepatitis (NASH), gout, rheumatoid arthritis, osteoarthritis, Crohn's disease, chronic obstructive pulmonary disease (COPD), chronic kidney disease (CKD), fibrosis, obesity, type 2 diabetes, multiple sclerosis, dermatological disease (e.g., acne) and neuroinflammation occurring in protein misfolding diseases (e.g., Prion diseases).

110. The method of any one of claims 106-108, wherein disease or disorder is a neurodegenerative disease; optionally, the disease or disorder is Parkinson's disease or Alzheimer's disease.

111. The method of any one of claims 106-108, wherein the disease or disorder is cancer; optionally, the cancer is metastasising cancer, brain cancer, gastrointestinal cancer, skin cancer, non-small-cell lung carcinoma, head and neck squamous cell carcinoma or colorectal adenocarcinoma.

112. The compound of any one of claims 1-99, or the pharmaceutical composition of any one of claims 102-104, for use in inhibiting inflammasome activity; optionally, the inflammasome is NLRP3 inflammasome, and the activity is *in vitro* or *in vivo*.

113. The compound of any one of claims 1-99, or the pharmaceutical composition of any one of claims 102-104, for use in treating or preventing a disease or disorder.

114. The compound or pharmaceutical composition of claim 113, wherein the disease or disorder is associated with an implicated inflammasome activity; optionally, the disease or disorder is a disease or disorder in which inflammasome activity is implicated.

115. The compound or pharmaceutical composition of claim 113 or claim 114, wherein the disease or disorder is an inflammatory disorder, an autoinflammatory disorder, an autoimmune disorder, a neurodegenerative disease, or cancer.

116. The compound or pharmaceutical composition of any one of claims 113-115, wherein the disease or disorder is an inflammatory disorder, an autoinflammatory disorder or an autoimmune disorder; optionally, the disease or disorder is selected from cryopyrin-associated autoinflammatory syndrome (CAPS; e.g., familial cold autoinflammatory syndrome (FCAS), Muckle-Wells syndrome (MWS), chronic infantile neurological cutaneous and articular (CINCA) syndrome/ neonatal-onset multisystem inflammatory disease (NOMID)), familial Mediterranean fever (FMF), nonalcoholic fatty liver disease (NAFLD), non-alcoholic steatohepatitis (NASH), gout, rheumatoid arthritis, osteoarthritis, Crohn's disease, chronic obstructive pulmonary disease (COPD), chronic kidney disease (CKD), fibrosis, obesity, type 2 diabetes, multiple sclerosis, dermatological diseases (e.g., acne) and neuroinflammation occurring in protein misfolding diseases (e.g., Prion diseases).

117. The compound or pharmaceutical composition of any one of claims 113-115, wherein disease or disorder is a neurodegenerative disease; optionally, the disease or disorder is Parkinson's disease or Alzheimer's disease.

118. The compound or pharmaceutical composition of any one of claims 113-115, wherein the disease or disorder is cancer; optionally, the cancer is metastasising cancer, gastrointestinal

cancer, brain cancer, skin cancer, non-small-cell lung carcinoma, head and neck squamous cell carcinoma or colorectal adenocarcinoma.

119. Use of the compound of any one of claims 1-99 or a pharmaceutically acceptable salt thereof in the manufacture of a medicament for inhibiting inflammasome activity; optionally, the inflammasome is NLRP3 inflammasome, and the activity is *in vitro* or *in vivo*.

120. Use of the compound of any one of claims 1-99 or a pharmaceutically acceptable salt thereof in the manufacture of a medicament for treating or preventing a disease or disorder.

121. The use of claim 120, wherein the disease or disorder is associated with an implicated inflammasome activity; optionally, the disease or disorder is a disease or disorder in which inflammasome activity is implicated.

122. The use of claim 120 or claim 121, wherein the disease or disorder is an inflammatory disorder, an autoinflammatory disorder, an autoimmune disorder, a neurodegenerative disease, or cancer.

123. The use of any one of claims 120-122, wherein the disease or disorder is an inflammatory disorder, an autoinflammatory disorder or an autoimmune disorder; optionally, the disease or disorder is selected from cryopyrin-associated autoinflammatory syndrome (CAPS; e.g., familial cold autoinflammatory syndrome (FCAS), Muckle-Wells syndrome (MWS), chronic infantile neurological cutaneous and articular (CINCA) syndrome/ neonatal-onset multisystem inflammatory disease (NOMID)), familial Mediterranean fever (FMF), nonalcoholic fatty liver disease (NAFLD), non-alcoholic steatohepatitis (NASH), gout, rheumatoid arthritis, osteoarthritis, Crohn's disease, chronic obstructive pulmonary disease (COPD), chronic kidney disease (CKD), fibrosis, obesity, type 2 diabetes, multiple sclerosis, dermatological diseases (e.g., acne) and neuroinflammation occurring in protein misfolding diseases (e.g., Prion diseases).

124. The use of any one of claims 120-122, wherein the disease or disorder is a neurodegenerative disease; optionally, the disease or disorder is Parkinson's disease or Alzheimer's disease.

125. The use of any one of claims 120-122, wherein the disease or disorder is a dermatological disease; optionally, the disease or disorder is acne.

126. The use of any one of claims 120-122, wherein the disease or disorder is cancer; optionally, the cancer is metastasising cancer, brain cancer, gastrointestinal cancer, skin cancer, non-small-cell lung carcinoma, head and neck squamous cell carcinoma or colorectal adenocarcinoma.